| INCH-POUND |

MIL-C-48394 (AR)
AMENDMENT 2
20 May 1990
SUPERSEDING
AMENDMENT 1
7 April 1976

MILITARY SPECIFICATION

CARTRIDGE, 81MM, SMOKE, WP, M375A3
ASSEMBLILNG, MARKING AND PACKING

This amendment forms a part of Military Specification MIL-C-48394 (PA) dated 6 November 1975, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department Defense.

PAGE 1

2.1, line 6: Specification "MIL-C-48156", delete "Cartridge, Ignition, XM299, LAP", and substitute, "Cartridge, Ignition, M299, LAP".

2.1, line 14: Delete "9230175 - Container.....in Jungle Wrap" and substitute "9276596 - Container, Ammunition, Fiber, M252A5 for 81MM Mortar Ammunition".

2.1, Add the following after last line on page;

"9287603 - Packing and Marking for Container, Fiber, M252A5, Ammunition for 81MM Mortar Ammunition in Jungle Wrap".

PAGE 2

* 2.1, Add under Inspection Equipment Drawing as follows:

"9280418 - Fin Distortion Gage."

PAGE 3

4.2.2, Examination, Add the following after the last sentence of the paragraph:

AMSC N/A FSC 1315

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"The contractor may elect to perform 100% inspection with automated equipment or impose mandatory process controls on automated operations in lieu of conducting sampling inspections for defects listed in this specification. In such cases, calibration plans shall be submitted to the Technical Agency (see 6.5) for approval for each automatic inspection station and process control plans must be sukbmitted for approval for each of the automated operations. The plans shall identify; (1) the frequency (production interval) for validating equipment functioning, (2) the defect standards and (3) a description of material control procedures to be imposed on product passing through each automated station during the production interval between valildations. Under these provisions final acceptance will be withheld from material produced during a production interval until satisfactory validation of each of the automated stations so controlled. In the event of a validation failure of any station, the material processed through such station since the last acceptable validation shall be placed in a rejected status pending corrective action".

PAGE 4

4.2.2.1, Change number in drawing box from "8835263" to "8885263".

PAGE 5

4.2.2.2, Delete "Major A" and substitute "Major", Delete "9294735" in Next Higher Assembly Block.

PAGE 6

4.2.2.3, Delete "Major A" and substitute "Major", Delete "9294735" in Next Higher Assembly Block.

PAGE 7

4.2.2.4, line 2: Delete "Critical 1. Ignition cartridge head not fully seated....Visual" and add "Critical 1. Depth of ignition cartridge from fin end below minimum...100%.....3.2....Gage CD".

* 4.2.2.4, delete Critical 3 in its entirety and substitute as follows:
"3. Propelling charge missing or damaged to the extent that propellant shall escape (if applicable, see 4.3.3.1)....100% inspection...3.2....Visual (see 4.3.3)."

* Delete Critical 5 in its entirety.

Add: "6. Cracks in fin housing...100% 3.2....Visual (4 power magnification)".

* 4.2.2.4, above Major A add the following "Special a....WARNING notice(s) missing....100% 3.2.....Visual".

4.2.2.4, Delete "Major A" and substitute "Major", Delete "9294735" in Next Higher Assembly Block.

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* Add new Major 110 as follows:

"110 Propelling charges damaged but not to the extent that propellant shall escape (if applicable, see 4.3.3.1)...100% inspection...3.2...Visual (see 4.3.3)."

* Add new Major 111 as follows:

"111 Fin blade damaged or distorted....100% inspection...3.2...Gage (see dwg. 9280418)"

PAGE 8

4.2.2.4, Delete "Major A" and add "Major", Delete "9294735" in Next Higher Assembly Block.

4.2.2.4, line 14 from top of page: Delete "202. Pettman cement does not cover....Visual" in its entirety and substitute "Minor: 202. Pettman cement (or alternate) improperly applied (applicable to the fuze, fin and ignition cartridge ....0.65%....3.2....Visual".

PAGE 9

4.2.2.4, line 2: Delete this line in its entirety.

4.2.2.4, Delete "9294735" in Next Higher Assembly Block.

PAGE 10

4.2.2.5, Drawing Number: Delete "9230175" and add "9276596".

4.2.2.5, Next Higher Assembly: Delete "9230175" and add "9287603".

* 4.2.2.5, Add Special A as follows:

"Special A Propelling charge support missing....100% inspection....3.2 ...Visual"

Add new Critical 1 and Note (a) as follows:

"1. Propelling charge damaged but not to the extent that propellant shall escape...25(a)...Visual (see 4.3.3)"

Add Major 101 as follows:

"101. Propelling charge damaged but not to the extent that propellant shall escape...25(a)...Visual (see 4.3.3)"

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Add Note (a) as follows:

"(a) Twenty five cartridges which have the support for the propelling charges assembled shall have the support removed and remove the four propelling charges. The propelling charges shall be inspected for damage (splits, ctacks, tears) which may result from assembling/disassembling of charges and support on fin boom. The inspection plan for Major defects shall be 100-1-2."

PAGE 11

4.2.2.6, Drawing Number: Delete "9230175" and add "9287603".

4.2.2.6, Next Higher Assembly: Delete "9230175" and add "9287601".

4.2.2.6, Delete "Major A" and substitute "Major"

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4.2.2.7, Drawing Number: Delete "9230175" and add "9287603".

4.2.2.7, Next Higher Assembly: Delete "9230175" and add "9287601".

4.2.2.7, Delete "Major B" and add "Major".

4.2.2.7, In catagory column: Delete "131 and 132" and substitute "101 and 102" respectively.

4.2.2.7, Following line 7: add "203...Kraft paper missing or not properly assembled...0.65%....3.2....Visual".

PAGE 13

4.2.2.8, line 2: Delete "Boards split or separated" and substitute "Boards split or separated exposing container".

4.2.2.8, Change number in Next Higher Assembly Block form "9278234" to "9287601".

* Add new page 13a.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.2.2.9	Packing and Marking for Box, wirebound Packing, Ammunition for 81MM Mortar Ammunition in Jungle Wrap		1 OF 1		9287601
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor 201	Car seal missing, unsealed or improperly positioned		0.65%	3.2	Visual

NOTES:

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PAGE 14

* Add new paragraph 4.2.2.10 as follows:

"4.2.2.10 Support (see drawing 9369933)

Catagory	Examination	AQL	Method
Major			
101	Inside length	0.40%	Gage
102	Rib Radius	0.40%	Gage
103	Inner Radius (small)	0.40%	Gage
104	Length between open end	0.40%	Gage

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* 4.3.1, line 4: Add as follows: "The cartridges shall be radiographically examined for proper WP void position prior to but within two hours of the actual test firings. The radiographs are for information only and will only be developed and forwarded to Picatinny Arsenal, ATTN: SMCAR-FSA-IM when the lot fails the acceptance requirement for horizontal range standard deviation or duds."

4.3.1, line 5: Delete "One Half of the sample cartridges shall be fired at charge one and other half at charge nine." and add "One half of the sample cartridges shall be fired at charge zero and the other half at charge four."

4.3.1, line 8: Delete "The individual velocities, pressures and ranges shall be recorded on the firing record" and substitute: "The individual velocities, pressures (charge four only), deflections and ranges shall be recorded on the firing record".

* 4.3.1, line 10: Delete "In the event of a misfire, the ignition cartridge shall be removed and forwarded to Picatinny Arsenal" and substitute: "In the event of a misfire, the ignition cartridge shall be removed and forwarded to Picatinny Arsenal, ATTN: SMCAR-FSA-IM".

PAGE 16

4.3.1, line 16 from top of page: Delete "...and prints of all the other rounds in that particular charge zone, shall be attached to the firing record." and add "...and prints of all other rounds in that particular charge zone, in that lot, shall be attached to the firing record".

* Add new paragraph 4.3.3 as follows:

* 4.3.3 Damaged propelling charge. To ascertain that a damaged propelling charge is to be classified as critical or major, the following procedure shall be used:

a. Place the damaged charge on a M24 fin, oriented such that the propellant would be resting against the damaged surface.

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b. The fin-charge shall be hand-shaken vigorously three times.

c. The examination shall be made for propellant spillage - if any propellant is observed, the damaged propelling charge shall be classed as a critical defect; if no propellant is observed, the damaged propelling charge shall be classed as a major defect."

"4.3.3.1 Alternate assembly/inspection procedure. The contractor may elect to utilize the following assembly operations:

a. Place the support for the propelling charge in a holding fixture.

b. Properly place the four propelling charges in the support with gaps aligned upward.

c. Press the fin housing into the propelling charges and support assembly (step b).

d. 100% visual inspection for damage to the ends of four propelling charges will be performed.

e. Continue the assembly/inspection operations (see 4.2.2.5)."

5.2 Delete this paragraph in its entirety and substitute: "5.2 Packaging, packing and marking - Packaging, packing and marking shall be in accordance with dwg. 9287603, 9287601, 9276596".

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* 6.2, line 3: Delete this line in its entirety and add: "ARDEC, ATTN: AMSMC-QAT-I, Picatinny Arsenal, New Jersey 07806-5000."

* Add new paragraph 6.5 as follows:

"6.5 Submission of Calibration and Process Control Plans. Prior to implementation of calibration or process control plans in accordance with 4.2.2, plans shall be approved by ARDEC, ATTN; AMSMC-QAT-I, Picatinny Arsenal, New Jersey, 07806-5000".

* Add new paragraph 6.6 as follows:

"6.6 Special defect. A defect, other than critical, that judgement and experience indicate may, depending upon the degree of variance from the design requirement:

a. Result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product or

b. prevent performance of the tactical function of a major end item."

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